

Treat the following exercises:

Exercise I: (3 pts)

Consider the **class diagram** for each of the following sentences:

- a) A country has a capital.
- b) A file is an ordinary file or a directory file.
- c) A file contains records.
- d) A class may have several attributes.
- e) A relation could be an association or a generalization.
- f) A polygon is composed of an ordered set of points.

Exercise II: (3 pts)

Create an **activity diagram** for the use case « Create a repair sheet ». To create a repair sheet, the workshop foreman enters the search criteria of the cars in the system. The repairs management software gives him the list of cars that match the criteria entered. If the car exists, the foreman will select the car. The software will then provide information on the vehicle. If the car is under warranty, the foreman must enter the date of reparation order. If the car does not exist, the foreman will enter the information about this new vehicle. In all cases, the foreman will enter the receiving and the returning date. If the damage of the car is paid by the insurance, the software will provide a list of insurances to the foreman. He will select the appropriate insurance. Finally, the software saves the repair sheet.

Exercise III: (3 pts)

We want to manage the different objects that contribute to the activity of a flowers sale store.

- The customer asks the seller some information about floral composition.
- The seller provides him all the necessary information.
- Then the customer orders the composition of his choice and the seller prepares the manufacture order and transmits it to his working florist.
- Then the seller edits the corresponding invoice.
- The florist creates the composition then archives the manufacture order.
- The florist hands the composition to the seller.
- The invoice is handed to the customer for payment once the bouquet is realized.
- Once the invoice is paid, the customer takes his composition and leaves the store.

Design this situation by a sequence diagram.

Exercise IV: (3 pts)

A chess game can be described as follows:

- Initially, the turn to play is for the player with the white pieces.
- When the player with white pieces, plays the turn to play is for who has the black pieces. The reverse is also true.
- When the player with the white pieces (respectively, the black pieces) does a checkmate, he wins the game.
- If there is a stalemate (the king cannot move without placing it in check), the game ends in a draw.

Imagine that there is a controller that monitors the chess game. Describe, from the point of view of the controller, the **state diagram** of the chess game.

Exercise V: (4 pts)

In school, we want to manage the booking of classrooms and teaching materials (laptop or / and Video projector). Only the teachers are empowered to make reservations (subject to availability of the room or equipment). The schedule of the rooms can be accessed by everyone (students and teachers). On the other hand, the summary scheduled by teacher (calculated from the planning of rooms) could be accessed only by teachers. Finally, there is for each training, a teacher who can only edit the Summary schedule for all of the training.

Give the corresponding **use case diagram**.

Exercise VI: (4 pts)

A gardener performs two types of works: the watering and hoeing. Watering is watering plants (tulips, eucalyptus or geraniums) with a tool (watering can or hose) containing water and the hoeing is to turn the soil with a tool (shovel or pickaxe) to fertilize it. In other words, the gardener uses a tool (watering, hose, shovel or Pickaxe) to put a resource (water or fertilizer) on a natural object (land or plant); this is produced by a work (watering or hoeing).

1) Draw the **class diagram**.

2) Draw an **objects diagram** corresponding to the following text:

Jacques is a gardener watering 3 geraniums with a watering can filled with water.

Jules is a gardener hoeing the Earth with a shovel to put fertilizer.